

# Additional Topic – Angles of Elevation and Depression

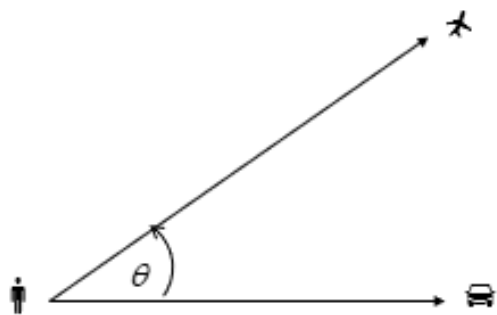
## Key Idea – Angles of Elevation and Depression

---

An **angle of elevation** is an angle measured from the **horizontal** in an **upwards direction**.

### *Example:*

A man is standing on level ground 200 metres from a car and observes a plane flying 500 metres above the car. Calculate, correct to the nearest minute, the **angle of elevation** of the **plane** from the **man**.



$$\tan \theta = \frac{o}{a} = \frac{500}{200} = 2.5$$

$$\theta = \tan^{-1} 2.5$$

$$= 68^{\circ}12'$$

### Exercise:

A fisherman is sitting in a boat 100 metres from the base of a 40-metre-high cliff. On top of the cliff is a 20-metre-high lighthouse. Calculate the angle of elevation from where the man is sitting to the top of the lighthouse.

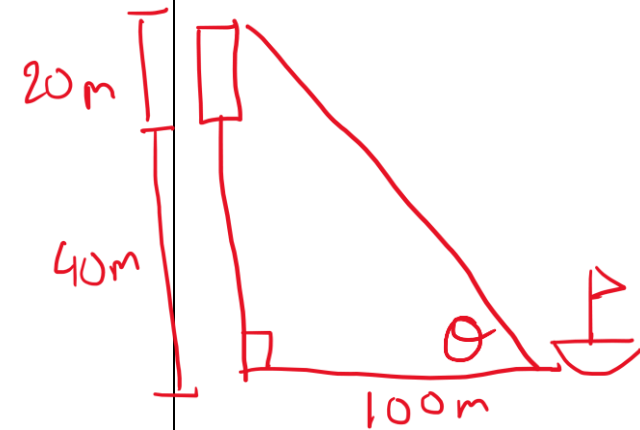
find  $\theta$

$$\tan \theta = \frac{40+20}{100}$$

$$\theta = \tan^{-1}\left(\frac{60}{100}\right)$$

$$= \tan^{-1}\left(\frac{3}{5}\right)$$

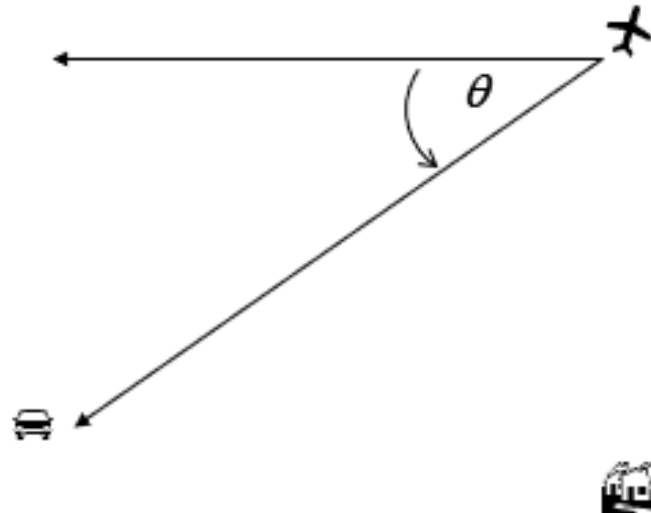
$$= 30^{\circ}57'50''$$



An **angle of depression** is an angle measured from the **horizontal in a downwards direction**.

*Example:*

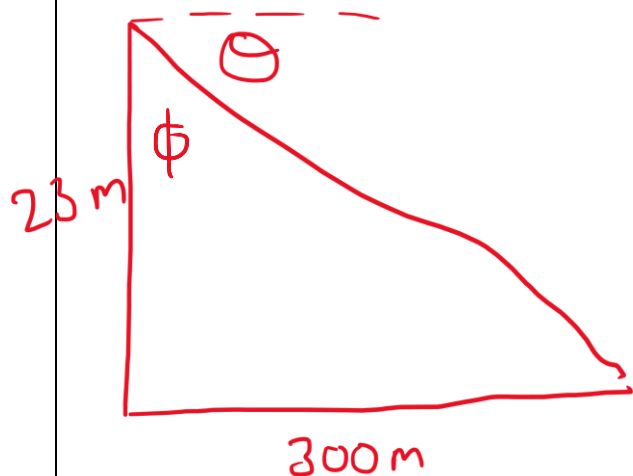
A plane is flying 2km above a house located 500m from a road. Calculate, correct to the nearest minute, the **angle of depression** of the **road** from the **plane**.



$$\begin{aligned}\tan \theta &= \frac{2000}{500} \\ \theta &= \tan^{-1} 4 \\ &= 75^{\circ}58'\end{aligned}$$

### Exercise:

A woman standing on the edge of a cliff, which is 23 metres high, observes a ship at sea 300 metres from the base of the cliff. Calculate the angle of depression from the woman to the ship.



find  $\Theta$

• first find  $\phi$

$$\tan \phi = \frac{300}{23}$$

$$\phi = \tan^{-1}\left(\frac{300}{23}\right)$$
$$= 85^{\circ}36'57''$$

$$\Theta = 90 - \phi$$
$$= 90 - 85^{\circ}36'57''$$
$$= 4^{\circ}23'03''$$